

Second Edition

RESEARCH METHODOLOGY IN APPLIED ECONOMICS

Organizing, Planning, and Conducting
Economic Research

Don Ethridge



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Planning the Research

Previous chapters provided general perspective and understanding about research in economics—what research is, what it can and cannot do, its role and purpose in the quest for knowledge or understanding, its relationship to and dependence on science, and how philosophical perspectives influence research activities. This understanding is useful and necessary for those engaged in research activities to comprehend and appreciate how individual research activities and efforts contribute to the general body of reliable knowledge.

But the foregoing chapters do little to provide guidance on how to *do* research. The generalizations about types of research and philosophies, when taken in isolation, probably constitute necessary, but not sufficient, understanding for those attempting to become economic research practitioners. What do I do? Where do I start? These questions and the perceived need to provide useful guidance for students were the reasons that this book was undertaken. This chapter and subsequent chapters are written with a view toward students' perspectives and needs as they endeavor to engage in research. Our system of higher education tends to "feed" students their training and understanding (the mixture of training and understanding varies) in somewhat self-contained subject-matter packages. The system appears to be effective as evidenced by the United States being recognized throughout the world for its superiority in graduate education. However, for students to make the transition from absorbing what is already known to using what is already known to produce *more* information or knowledge, we leave them to their own devices except for the individualized assistance (apprenticeship) from their teachers, graduate advisors, and committees. The contention is that both the students and their advisors can benefit from some prescriptive guidelines that help to provide a structure, yet are flexible enough to facilitate individual creativity and the circumstances and needs of particular research efforts. Again, this and subsequent chapters are heavily focused on the perceived needs of students. In the larger sense, we are all students; some have just been at it longer than others.

The material in this part of the book is not particularly unique or insightful when considered with hindsight of experience. It is time tested and very dependable. It is deceptively simple as a concept or approach, yet inherently difficult and taxing in implementing as a specific project. As a general approach to the process,

it will rarely, if ever, fail. It is *the* approach to problem (or issue or question) solving in economic science, and to a large extent works for problem solving of any kind, be the problem disciplinary, subject-matter, or problem-solving. That is, the approach of identifying and understanding the problem and objectives, then designing an approach to accomplish those objectives is a reliable formula for success with all kinds of problems—research, policy, management, personal, and so on.

This part of the book—chapters 5 through 9—will make frequent references to appendices A, B, and C. These appendices consist of example research proposals. Their purpose is to help illustrate the principles and guidelines prescribed in the procedural part of the book. The research proposal in appendix A was selected to represent a proposal prepared for a government agency. The proposal in appendix B is a graduate student thesis proposal for an M.A. program in economics. The proposal in appendix C is for a Ph.D. dissertation. These proposals will be referenced to illustrate, and sometimes contrast, certain procedural points. The proposals as a group help to show, among other things, both similarities and differences in structure, approach, and sophistication in research proposals.

The first section of this chapter discusses the research proposal as the planning vehicle, followed by a section that describes the common components of research project proposals. The third section covers some of the considerations in evaluating proposals, followed by a brief discussion of effective use of the proposal and some perspective of public support for economic research. The final two sections address the importance of writing in planning, conducting, and reporting research and some guidelines on scientific and technical writing in economic research.

THE RESEARCH PROJECT PROPOSAL

The key to successful research is the *research plan*. Recall that a central element of the definition of research is that it is a systematic approach (chapter 2). In order to target understanding or knowledge on specific matters, be they disciplinary, subject-matter, or problem-solving oriented, the approach needs to be carefully constructed and designed to ensure that the information or knowledge produced by the activity is reliable. To derive the knowledge without a design occurs only by accident. This does not say that it is impossible to discover public reliable knowledge by accident: that is, by simply pursuing idle curiosity. But accidental discovery happens in a small proportion of cases and is not dependable—like relying on a random event to deliver the understanding. The heart of the research plan is the *research project proposal*.

The intent is not to denigrate the importance of accidental discovery. Being open to new thoughts, ideas, approaches, and insights is an integral part of creativity in the research process. But the most insightful accidental discovery in economic research usually occurs within the boundaries of a structured direction

of inquiry. The research proposal should not limit creativity, but rather channel it toward objectives the researcher has discerned through a reasoning process.

Research proposals are generally required for any type of research endeavor and by all entities that support research. Specific requirements for proposals vary drastically. Research proposals that seek financial support differ from research proposals that serve only as the guide for the researcher. The most stringent requirements in terms of completeness and exactness may be for Ph.D. dissertation proposals, although some federal agencies, or some types of federal research funding programs, require elaborate, more highly planned and detailed proposals (see appendix A). Other funding sources, for example, some industry groups, may prefer proposals that are concise, brief, and to the point; they may wish to forego conceptual and other "academic" aspects of the research in the proposal. In rare cases the proposals may be oral instead of written, although when oral proposals are required, they usually are in addition to a written proposal.

Irrespective of the specific requirements of the proposal, it constitutes evidence of the research plan. The more thorough and clear the proposal, the clearer and more complete the plan. The importance of effective written communication skills is difficult to overstate, not that communication skills will always produce good proposals, but they are impossible to produce without communication. There must first be clear, well-developed thoughts that are then communicated effectively. To whatever extent *both* are not present, the proposal will fail to achieve its purpose. When the proposal is not clear, the only reasonable assumption the reader can make is that the writer's thinking is not well developed. When someone says, "I understand it, I just don't know how to say it," it really means, "I don't understand it clearly enough to explain it."

The proposal thus serves dual purposes, one for the researcher and one for the evaluator(s) of the proposed research. For the researcher, it provides an operational plan for conducting the work while simultaneously keeping the researcher from wandering off on questions or issues that are, albeit worthy, not germane to the purpose of the study. It is also a vehicle that forces the researcher to more thoroughly understand the reasons for and intent of the research, to anticipate potential problems, and develop plans for dealing with them before they occur. For evaluators (e.g., administrators, funding individuals or groups, or graduate student advisory committees), the proposal is the means through which they determine the intent of the research and make approval/disapproval and funding decisions. It also provides the means for them to require accountability from the researcher.

ELEMENTS OF THE RESEARCH PROPOSAL

Although proposals vary in their complexity and configuration, there is a set of common parts or elements. These are identified and discussed briefly below. The listing is not exhaustive, but it is representative, and all proposals do not contain all elements. Figure 5.1 is used in conjunction with the following listing to help

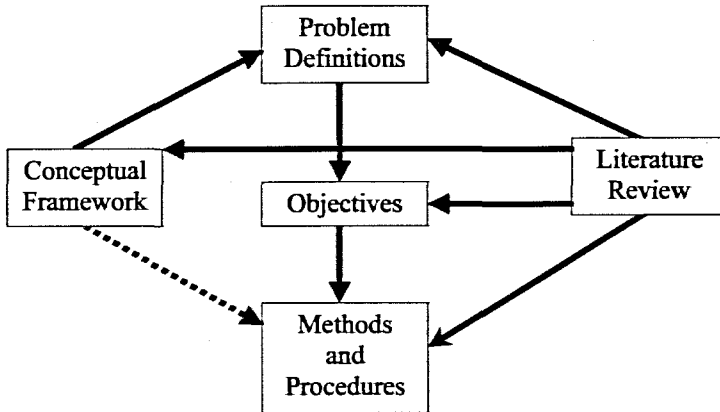


Figure 5.1. Linkages among components of a research project.

describe how some of the main working components of the proposal relate to other components. The list draws on the material of Williams (1984).

A. *Title.* The title of the project should be descriptive of the major focus of the research, but it should be no longer than necessary. The title cannot possibly convey a thorough understanding of details of the study, but it can provide an accurate impression of the central focus of it. Doing so in the most efficient manner (the least unnecessary words) is desirable. For example, a project title such as "An Economic Study of the Impacts of Lowering Import Tariffs on Textiles on Consumers, Textile Manufacturers, and Fiber Producers in the United States" can easily be shortened to "Lowering Import Tariffs on U.S. Textiles: Effects on Consumers and Industries." Both titles convey approximately the same understanding of the intent and content of the proposed research, but the latter does so with fewer words.

B. *Identifying information.* This tells something about the people and organizations that would be involved in the proposed research and other types of summary information relevant to the project. For example, the names, titles, addresses, and telephone numbers of project personnel may be given. The people may be divided into groups such as project leader (or principal investigator, etc.), research advisors or cooperators (individuals or organizations) and, in the case of graduate student proposals, graduate committee members (see appendices B and C). An indication of the budget for the research and specification of the time required for the research (dates for project initiation and completion) may be included. Additionally, a detailed budget, when needed, is usually presented elsewhere in the proposal. A detailed work plan may be included elsewhere, as well as biographical résumés of the research personnel. When résumés are provided, they are usually appended to the proposal rather than integrated into the body of

the proposal. The type, amount, and format of the identifying information may vary and should be developed with (1) instructions about format specifications and (2) the readers of the proposal in mind. Some examples of formats for identifying information are shown in appendices A through C, but there are many acceptable variations.

C. *Problem identification and explanation.* This part of the proposal describes the problem being addressed by the proposed research and provides the rationale for the proposed objectives (represented by the solid line between objectives and problem definition in figure 5.1). The problem may address disciplinary, subject-matter, or problem-solving issues. It is perhaps best described in a two-step procedure, with the first step developing the general perspective of a broad problem area and the second step narrowing attention to a carefully developed subpart of the general problem that is researchable within the resource constraints of a single project. The researchable problem is the reason (justification) for the proposed research. In some cases this section of a proposal may even be titled "Justification." In other cases, format may call for both a problem statement section and a separate justification section, but when that is the case the two sections are closely related. A problem statement or justification is, to the author's knowledge, a universal requirement for research project proposals. Chapter 6 takes up the matter of problem definition in further detail.

D. *Objectives.* The objectives, also universally required in research proposals, specify precisely what the proposed research project intends to find out, discover, and/or accomplish. Objectives identify the goals or ends of the research, not the means. Objectives are usually best stated as a one-sentence general objective and a list of specific objectives. They need to be stated clearly, but concisely, and oriented to producing or discovering knowledge or information, or perhaps prescriptions in subject-matter or problem-solving research. Objectives are justified by the problem statement and are the point of focus of the methods and procedures (figure 5.1). Further elaboration of these principles is also provided in chapter 6.

E. *Review of literature.* This part of the proposal provides a summary review of the research literature that is relevant to the study being proposed. Its nature and structure can vary. For example, it may be a separate section in the proposal or it may be integrated into other sections (problem, methods and procedures, or conceptual framework). The literature reviewed may be related to the proposed research through the problem, objectives, methods and procedures, and/or conceptual framework, as indicated by the solid lines to those components in figure 5.1, but it should be confined to scientific, as opposed to popular, literature. Its purpose is to provide the base of knowledge on what we have already learned about matters that are addressed in the proposed research. This knowledge base is for the benefit of both the researchers and the evaluators of the proposed research. Although the literature review is presented here after the objectives, in practice it is difficult, perhaps impossible, to fully develop the problem

statement and objectives prior to conducting a thorough literature review (although the written literature review does not necessarily have to be written at that point). All proposals do not require a formal literature review section, but no research project should be undertaken without the researchers being knowledgeable of the prior research related to the topic. The review of literature is the topic of chapter 7.

F. *Conceptual framework.* A formal conceptual framework is not a universal requirement for research proposals, but any economist who attempts to forego this step risks making potentially serious logical errors in his or her research. A formal conceptual framework should be standard in research proposals by students, if only to ensure their comprehension of the principles as they relate to the research problem. In its most basic form, the conceptual framework is an analysis, using economic, and perhaps other, concepts of the specific researchable problem that the proposed research addresses. *It is a conceptual analysis of the problem.* Its relationship to other parts of the proposal is illustrated in figure 5.1. It may be directly affected by the literature review. The conceptual framework also may lay the foundation for some of the research methods and procedures (the dashed line in figure 5.1), but that is a side benefit, not its central purpose. The central purpose is to ensure that the researchers and evaluators are examining the problem with appropriate concepts, thus helping to obtain a clear perspective of the research problem/issue/question. In disciplinary proposals for research that is primarily theoretical, the conceptual framework might be very brief, or even absent. In some cases the primary objective of the research may be to develop a conceptual framework. In disciplinary research proposals in which the problem is measurement, the conceptual framework might consist of a theoretical analysis of that measurement problem. In research with a problem-solving focus, the conceptual analysis would be of that type of problem. The conceptual framework is the subject of chapter 8.

G. *Methods and procedures.* The methods and procedures describe how the objectives will be achieved. The methods and procedures flow directly from the objectives, are affected directly by the literature review, and may be indirectly affected by the conceptual framework (figure 5.1). They address matters such as data to be generated or collected, analytical techniques to be used, sequencing of procedures, and derivations from empirical estimates. Inclusion of methods and procedures is universal in economics research proposals, although the detail and complexity may vary to fit the background and preferences of evaluators. For example, a proposal being reviewed by economists probably would need more detail and economic sophistication in its methods and procedures than a research proposal being reviewed by noneconomists. The methods and procedures in student research proposals should contain as much detail as the student can envision in the planning stage of the research project. Methods and procedures are addressed in detail in chapter 9.

H. *References.* The list of references provides the documentation of the sources used in the proposal. References include those sources that may have

been used in explaining, defining, or documenting the problem, those used in the review of literature, and any references used in the conceptual framework or the methods and procedures. The references and referencing are covered in chapter 7, in conjunction with the review of literature.

EVALUATING RESEARCH PROPOSALS

When a research proposal is assessed, the soundness of the research plan or design is being evaluated. The proposal is evidence of the quality of thought that has gone into the project, and the whole research plan (i.e., the desirability of the proposed research project) will be judged by the quality of the project proposal. This necessitates clear, incisive thinking and planning, and it needs skillful communication. The planning of the research and the writing of it (in short, the proposal) are usually the more difficult parts of the research task. The process of carrying through with the research plan becomes more a matter of "going through the steps." This is obviously a simplification because thought, judgment, and decisions also will have to be used in carrying through with the plan. However, on a comparative basis, the defensibility of the research rests on the plan. Well-executed procedures on a poor research plan still produce poor research. Evaluators of potential or proposed research projects are concerned with the plan.

In a presentation at a research conference, an administrator (Smith, 1983) provided a list of the criteria by which administrators of research activities and funding evaluate applied (nondisciplinary) research projects. The criteria are not specific to economic projects, but they generally show what administrators and other evaluators of applied research proposals consider. Smith's slightly modified list is provided below.

- A. Is the investigator interested in the problem? This is important because genuine interest provides motivation and fosters creativity.
- B. Is there a genuine lack of knowledge related to the problem?
- C. Is the research recognized as needed by people other than the investigators? The perceived need should extend beyond the disciplinary interests of the researchers to "users" of the research.
- D. Are the objectives:
 - 1. appropriate to the problem?
 - 2. attainable?
 - 3. observable or measurable?
 - 4. sufficiently specific?
- E. Does the investigator have sufficient resources (education, training, facilities, material, and funding) for the project?
- F. Does the proposal recognize appropriate constraints, both internal (i.e., constraints of the mission or role of the organization) and external (e.g., constraints imposed by social, economic, or political dictates)?

G. Does the proposal, as evidenced by background investigation, indicate that the research is likely to be productive?

H. Is the expected value of the research greater than the cost of obtaining results?

I. Has the study been designed in a manner such that results are as widely applicable (to as many people, situations, locations, etc.) as is feasible?

These or similar criteria are widely used by administrators who evaluate proposals for problem-solving and subject-matter research. Disciplinary research proposals require a modified set of criteria, but disciplinary research often does not go through a formal proposal review phase. When disciplinary research is proposed and reviewed, it is disciplinary research of known relevance (see Johnson, 1986b, ch. 10 and pp. 195–198 for elaboration on these points). However, the support for disciplinary research is limited, and most disciplinary research in economics is done without overt support or direct administration. These contentions are supported further in a subsequent section of this chapter.

IMPORTANCE OF FLEXIBILITY

While a primary function of the proposal is to provide boundaries and guidance for the research, it must not be viewed as a “road map” from which there can be no deviations. The plan may have to be altered as the research progresses. Occasional modification of an objective may be needed to improve the focus and usefulness of the project. Sometimes one discovers, or even develops, a new method or procedure that achieves the objectives more effectively than the method or procedure originally proposed. One often develops a more thorough perspective of the research problem and/or the conceptual framework as one learns more of the issues while carrying out the research. One often discovers more related literature than was found in the original literature search; this alone can sometimes cause other parts of the proposal to be modified. Proposal writing is an evolving process.

The important point to be made is that the project proposal is an asset in the research process, but it must not become a straightjacket. Do not view it as something that confines your research, but as something that keeps your attention and research efforts focused on the problem and objectives defined for the research. It keeps you from letting your attention stray too far from the point of the research. You want to be efficient in conducting your research, but you do not want to stifle creativity in the process. As you have new thoughts and perspectives and encounter new ideas, consider and evaluate them. Integrate them into your plan if they are warranted and feasible within the general objectives and resource constraints.

A guideline on making proposal revisions after a proposal has already been approved is that minor revisions may be introduced into the plan by the researcher without formal approval, but major revisions should seek approval from the

reviewers or approvers of the original proposal. This holds in any situation in which the proposal goes through external evaluation, such as proposals for funding and graduate research proposals.¹ For example, a research project funded by an agency or group would need approval of substantive modifications in any of its objectives but might only require notifying the agency or group of some changes in procedures. Graduate student research could need formal graduate committee approval if major changes in methods and procedures are proposed. Determination of whether formal approval is needed for a proposed change is obviously a matter of judgment. One guideline is always reliable: When in doubt, *ask*.

FUNDING FOR ECONOMIC RESEARCH

Unlike the case with the "hard" sciences, the public perceives little direct need for disciplinary research in economics. Consequently, there is little direct support for disciplinary research in economics. Users of economic research are willing to provide financial support for subject-matter and problem-solving economic research because they can visualize direct benefits from it. Economists are sometimes perplexed by the absence of recognized need for the basic research in economics when they see public institutions (government, universities, foundations, etc.) content to finance disciplinary research in chemistry or biology, for example, that has no known immediate application, but see no need for disciplinary research in economics.

Economists may have done too little to "sell" the need for disciplinary research in economics to the public, the business community, and government. Some disciplines have been able, by accident or design, to create the general understanding that their pool of basic knowledge must be kept growing for benefits to society to accrue. For example, the public seems to accept the premise that basic research in chemistry is necessary; the public is willing to more readily assume or accept that the knowledge gained will eventually provide benefits to the public at large through yet-to-be-discovered applications. Yet there seems to be no such general presumption about economics or any of the other social sciences. There may be more popular support for purely disciplinary activity in the arts than in the social sciences.

Disciplinary research in economics is supported to a small extent by public institutions such as the National Science Foundation, by some government units, by universities that allow or encourage faculty to pursue their own disciplinary interests, and by subject-matter and problem-solving research projects within which some disciplinary research is undertaken as a means of achieving the applied research objectives. We have no reliable information regarding the proportion of support from these types of sources, but their relative importance is probably in reverse order of the above listing. The point is that even disciplinary research in economics tends to be in response to perceived needs to address current issues

and problems. The disciplinary research in economics is typically directed to the inability of economic theory or procedures to provide adequate insight and guidelines for issues of the day. John Maynard Keynes's *General Theory of Employment, Interest, and Money* was in response to the failure of the formal economic theory of the 1920s to provide guidelines on how to deal with the problems of a depression economy. Johnson (1986b, pp. 121–138) provides an analysis of two cases of economic research that also supports this position—Wassily Leontief's research on input/output relationships and T. W. Schultz's work on human capital.

There is, however, public support for applied problem-solving and subject-matter economic research. Public administrators, elected officials, business leaders, and even a portion of the general public grasp the need to understand and predict consequences of economic actions and phenomena and economic effects of social and political actions, to generate prescriptive solutions to management and policy questions, and such matters. These groups are interested in positivistic knowledge, normativistic knowledge of values, and conditionally prescriptive knowledge. However, the mind-set is that they want the research to generate the knowledge to be directed to specific, defined problems as opposed to general, conceptual probing for knowledge. Our external clientele take a highly pragmatic interest in our research.

Economists have concerns about public support, financial and other, for economic research. Readers interested in the author's perspective on research funding in economics, including some suggestions for young professional economists, may want to read appendix E.

An offshoot of this discussion is that economists cannot be easily grouped into theoretical economists on the one hand and applied economists on the other. The *relative* emphasis we place on theoretical or conceptual components of the applications may, and do, differ, but we all focus on current issues with which public and private decision makers are concerned. Furthermore, economists who are conducting multidisciplinary subject-matter and problem-solving research cannot conduct that research without also being capable theoreticians; they must grasp the theoretical aspects of the problems they address. However, with applied (multidisciplinary) research, understanding the problem theoretically is not sufficient. Conducting applied research requires the economist to go beyond the confines of the theory (recall that all theory is to some degree a generalization and simplification of the real world). Applied economic research requires that we use the theory to provide a framework for analysis, but the approach also must recognize and deal with other circumstances that must be accounted for and that may modify the theoretical outcome.

IMPORTANCE OF WRITING

"To write well is to think well" (Ghebremedhin and Tweeten, 1994, p. 91). Even if good writing is not, in itself, good thinking, it is the *evidence* of good think-

ing. It has also been called “visible thinking” (Mighell and Lane, 1973, p. 15). There is a degree of alarm within educational circles, at least within the United States, about the capacity of people in general to write well. The reasons are varied and only vaguely identified, but it has serious implications for research in general and economic research in particular. It is recognized as a problem in graduate programs in economics. Academic and nonacademic employers of economists with new graduate degrees, as well as journal editors and readers, have been vocal in their complaints about their level of writing and other communication abilities, especially their basic expository skills (Krueger et al., 1991, pp. 1048–1049). Even professional economists (e.g., Freedman, 1993) lament that economists (as well as other science groups) generally fail to communicate well enough to be understood because of disciplinary chauvinism, excessive use of jargon, including mathematics, and a more general failure to write well (clearly).

Writing is a skill that can be learned and taught. Some people obviously find writing easier than others, but few find technical or scientific writing an easy task. This may be largely due to the complex nature of the subject matter and the need to be very exact—leaving as little chance for misinterpretation as possible—in the communication. Yet economists should be at least adequate to the task because they must do a substantial amount of reading, which itself fosters the capacity to write. An essential part of developing the skill is to practice writing and to develop a productive, realistic attitude about it. Writing is not an activity that one decides to do, sets down and does it, and is finished. It is a *process*, and the process involves *rewriting*, usually numerous times. Effective communication is important to the research process in several ways, and investment in the time to practice, study, and develop the skill is both productive and necessary. But it is also often demanding, sometimes frustrating, and occasionally rewarding. The author sometimes, in an attempt at humor, tells students, when they are particularly frustrated with the demands of writing, to be encouraged—writing papers gets much easier after you have written two or three hundred of them. It is not always taken with the humor intended.

The importance of writing goes well beyond the function of reporting research, although that is one aspect of its relevance. Writing is the evidence of research productivity. Writing makes research results accessible to others. Writing affects one’s professional standing. The writing activity may be more critical for economics than for some other disciplines. But writing is a part of the research process itself (Ladd, 1987, pp. 77–78; Mighell and Lane, 1973, pp. 15–16).

In the process of developing effective written communication, we force ourselves to clarify and refine our thoughts, to place them in logical order, and to express them in ways that enable others to grasp our thought process and reasoning. Few of us fully understand the intricacies of our own thoughts on a matter until we have gone through the disciplined process of explaining them clearly in

written form. That is why the *written* research proposal is of premier importance in planning effective economic research. Ladd (1987, p. 79) further points out that writing plays different roles at different stages of the research process. For example, writing in very early stages may entail little more than scribbling notes but require painstaking precision and organization in preparation of final reports. For students, implementation of any of the planned methods and procedures should not be done before a written research proposal is completed and successfully defended.

WRITING GUIDELINES AND TIPS

The purpose of this section is to provide some general insights, suggestions, and admonishments about technical or scientific writing as it applies to economic research. The reader should note that these apply to all technical writing (e.g., research reports, journal articles), not just proposals. Note that this book is not written in technical style; the intent of this book is different from the purpose of a technical proposal or research report. None of the material in this section is unique to any particular source or author. Different aspects of this material can be found in various sources. Two specific references that are both useful and concise are Ghebremedhin and Tweeten (1994, pp. 90–100) and Mighell and Lane (1973, pp. 16–20).

The goal of writing in research is to convey information. Good writing is writing that serves its intended purpose. Research writing does not have to be dull (it should hold readers' interest), but the research audience is seeking information, insight, knowledge, and/or mental and intellectual stimulation rather than entertainment as recreation or diversion. You must assume that your audience wishes to understand and be informed as effectively and efficiently as possible, while not being bored or lulled to sleep in the process.

One recommended element in conveying information in research writing is to determine the intended audience and their information and knowledge needs. Knowing your audience includes an understanding of readers' levels of technical knowledge as well as their interests. A research proposal or report that is to be read by a group of government or industry policy makers must be written differently, with different aspects of the concepts, procedures, and results emphasized, than a report on the same research presented to a group of economists. Case in point: Economics journal articles are written to be read by economists. Hardly anyone except professional economists will even attempt to read a disciplinary or subject-matter economics journal. When noneconomists (and some economists, too) attempt it, they tend to be confused and "turned off" by the experience. A useful technique for some writers is to try to place themselves mentally in the anticipated readers' position as a means of focusing on their interests and prior knowledge.

Another reliable guideline is to *organize* what you have to say. You are trying to achieve a logical order of presentation of the thoughts and information that is clear, direct, and precise. You want a logical sequencing of thoughts, ideas, and information; you are “directing” your readers’ thinking and understanding. Achieving this logical order invariably requires an outline of what you are to present. The outline is the tool for organizing the flow—major points, subpoints, linking points to one another. It is advisable to construct a written outline (the more detailed the better) rather than trust an outline to memory.

Communication of logical order of thought and information is fostered by several characteristics that can improve your research writing. One is simplicity. Simple, direct statements communicate more effectively than do complex, rambling, or wordy statements. Use simple statements and phrases and avoid unnecessary qualifiers, jargon, clichés, and platitudes. For example, the sentence “While the estimates of elasticity of exports with respect to export prices made by various researchers show a large range of variation, that being from -0.3 to -1.2 , the predominance of the estimates falls within the range of -0.5 to -0.8 .” might be condensed to, “Prior export price elasticity estimates range from -0.3 to -1.2 , but are mostly in the narrower range of -0.5 to -0.8 .” Of course, the final determination must be consistent with the context.

Another desirable characteristic is precision (lack of ambiguity) in language. Sometimes a minor difference in the choice of words can alter the readers’ perception of the meaning of a statement. How often have you seen a writer, occasionally even a professional economist, use a phrase such as “demand increased” when he or she really meant “consumption increased” or “quantity bought increased?” “Demand increased” describes a shift in a demand relationship, whereas the other phrases signify either a shift in a demand relationship or a movement along a demand relationship. The ambiguity or lack of precision on such a fundamental distinction is a grave error for an economist. While ambiguity detracts from communication, we should recognize that simplicity and precision may conflict on occasion and necessitate a trade-off. This may be a problem, especially when the audience contains noneconomists. Technical terms with specific meanings sometimes must be foregone to achieve communication with noneconomist readers. The best advice on these situations is that they require judgments on a case-by-case basis.

Another guideline is to avoid advocacy and judgmental words and phrases. Keep the tone of the writing as impartial as possible. You are not an advocate for any group or position, and it is desirable to avoid any such connotation or appearance. Avoid personally judgmental words (e.g., *should*, *good*, *bad*, etc.). Beware of adjectives such as “unfortunately” or “regrettably” in descriptive phrases. Do not use “poverty-stricken people” when you can be both more precise and impartial with a phrase such as “families with incomes below the government-defined poverty line of \$4,000 per capita per year.” In short, be as complete, accurate, and objective in your writing attitude as you can.

An essential component of good scientific writing is having what you write reviewed, evaluated, and critiqued by persons who have experience and perspective in the same type of writing. For students, this should include evaluations and assistance from professors. None of us can be as unattached and analytical with our own writing as with someone else's writing. You want reviewers who are critical but constructive, so as to help you identify the weaknesses (and strengths) in the drafts you write. Remember that the service others provide in reviewing your written material is likely to require reciprocation. That reciprocity may, however, also benefit your own writing ability, as indicated below.

A practice that will help you become a more effective research writer is to critique other people's written material. By studying others' written material with a view toward being critical in a constructive way,² you develop a greater ability to be more objective and critical toward your own writing. For that reason, requiring students to critique each other's drafts of parts of their proposals and research reports (e.g., theses, dissertations, term papers, articles being prepared for submission to journals, etc.) is a useful practice. Written guidelines for doing critiques are provided in appendix D. Another useful practice that fosters effective writing is to reevaluate the writing after it has been set aside for a time. Write a draft, then return to it later, sometimes days or weeks later, with a fresh perspective.

"Style" is an important part of good writing. Each of us has our own style of writing and there is no single appropriate style. We also must recognize that writing style requirements may vary with the type of publication; always check on any mandated style requirements for any type of outlet. Most journals have some individual style preferences, different government agencies have specific style dictates, and most graduate schools have style guidelines for theses and dissertations. There are also general writing guides, such as *The Little, Brown Handbook* (Fowler, 1983) and *The Craft of Scientific Writing* (Alley, 1987), that provide guidance on matters ranging from punctuation and grammar rules to creating the mood for writing. Another type of useful aid is computer word processing editing software. In developing your own style, it helps to read good writers; it may help you to identify what you are comfortable with and what you like.

There are, however, some general style guidelines irrespective of your individual style preferences. These include avoiding complex phrases and sentences, making sure you have correct spelling (a much easier task with spell check options in word processing software) and punctuation, and avoiding digressions into topics not essential to the purpose at hand. A style technique that is useful in practically all cases is to begin by telling readers what you are going to do and summarize it for them at the end. There are also some technical points of style that may vary, but you may need to be aware of them. These include:

A. *Page margins and line spacings.* These vary with the outlet; unless specified, 1.25-inch top, bottom, and side page margins and double spaced lines are usually safe.

B. *Mathematical and statistical notation.* Some outlets promote technical symbols and notation while others prefer to avoid them (this is a consideration independent of the audience to which the writing is directed).

C. *Citations and reference notes.* Some outlets allow footnotes while others wish to avoid them. Endnotes are an option. Styles of referencing may vary (more on this in chapter 7). Whatever style you adopt, be consistent in its use throughout a written document.

D. *Tables, charts, and graphs.* "A good picture may be worth 10,000 words but a bad picture is worth hardly anything" (Mighell and Lane, 1973, p. 18). Style dictates vary greatly. Most table formats specify no vertical rules and the placement of table titles above the body of the tables. Standard format for charts is for the title at the top as well. Figure titles are standard at the bottom of the graph. However, many current graphics software packages force titles at the top, so this could change as a matter of expediency. A generally accepted rule of tables, charts, and graphs is that they should "stand alone." Titles should be explanatory; they should be presented such that readers can read and understand them independent of the text. All units should be specified. All columns, axes, and so on should be clearly labeled. Footnotes may be used to explain needed details or caveats. Complete reference citations should be provided with the visual if references are needed for sources or other purposes.

E. *Verb tense.* When to use present, past, or future tense can be confusing. In general, be wary of future tense except when writing the methods and procedures section of a research proposal. Preferences vary between present and past tense on proposals and reports. The most reliable rule of thumb, however, is to try to determine the preference of the targeted audience. It is generally better not to shift verb tense within a major section of a written document, although switching between major sections may be warranted.

F. *Personal pronoun form.* In formal technical writing, the rule historically has been to write in third person (e.g., he, she, it, they), not in first or second person (e.g., I, we, you). Journals in particular in recent years are deviating from this form. A safe general guideline is to use third person for formal research reporting, but first or second person is appropriate for less formal style such as presentations and proceedings papers. Consult specific guidelines.

Writing is, to a degree, a personal matter with most of us. We want our writing, as well as our oral communication, to be well received. That "need" is perhaps fostered, even instilled, by our early studies of expository writing; our literature and exposition teachers value "inspirational," "moving," "persuasive" writing that stirs the emotions. Some of these attributes are also valued in technical and scientific writing, but precision of meaning, accuracy, and analytical rigor are more important attributes.

Our consideration of writing and its role in research is not finished. The matter will be a point of attention at several junctures in subsequent chapters.

SUMMARY

This is the first of the book's six procedural chapters. The primary focus of the chapter is an overview of the research plan, embodied in the research proposal, with additional consideration of two closely related matters—planning and justifying research based on proposals and communicating in writing.

The written research proposal constitutes the research plan. Research project proposals may differ substantially with the type and purpose of research and the audience to which the proposal is directed. In general, the proposal is necessary for both the researcher (to organize the research activity and/or obtain support, financial and otherwise, for the research) and the evaluators charged with evaluating, approving, funding, and/or supervising the research activity. There are common elements for research project proposals—title, identifying information, problem description, specification of objectives, literature review, conceptual framework, methods and procedures, and references. All elements are not necessarily required in every proposal and the manner they are included may vary, but they constitute the standard set of elements.

Guidelines are provided on evaluation criteria and on modifying the research plan. The different ways in which economic research is funded are discussed and attitudes and support for the different types of economic research are considered.

Written communication within the context of both proposal writing and scientific writing in general is emphasized. Writing represents evidence of the thought process. It should be recognized as a process, involving rewriting, and as an integral part of the research process rather than an addendum to it. Some general suggestions for effective writing are provided, recognizing that there is substantial flexibility for individual style within scientific writing. Reviewing and critiquing are advocated as means of accomplishing effective writing.

RECOMMENDED READING

Alley. Chapters 1–13, 18–21, and the appendix, *The Craft of Scientific Writing*.
Ghebremedhin and Tweeten. *Research Methods and Communication in the Social Sciences*.
Ladd. *Imagination in Research*, pp. 77–81.

SUGGESTED EXERCISES

1. Locate three economic research proposals, one written by a graduate student for graduate program research, one submitted to a government agency for funding, and one to an industry group for funding. Describe how and to what degree the proposals differ in their components, detail, and organization.

2. Provide a written constructive critique of the graduate student proposal from the previous exercise.
3. Write a 1–2-page description of the differences between technical and expository writing.

NOTES

1. When the proposal's only function is to formulate the plan for the researcher (i.e., there is no external review), there is obviously no need for external approval for changes.
2. Being critically constructive may consist of a two-step procedure: (1) a critical reading to identify what is right/wrong, clear/confusing, effective/ineffective, and the like; then (2) correcting or making suggestions for changes.